

Question Paper Preview

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

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Corporate Finance

Section Id :	640653124711
Number of Questions :	31
Section Marks :	100
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531863634 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : CORPORATE FINANCE (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536037109. ✓ YES

6406536037110. ✗ NO

Question Number : 2 Question Id : 6406531863635 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A firm is considering an investment opportunity promising to pay Rs 10,000 in one year and Rs 12,000 in two years. If the prevailing constant rate of return is 8%, what is the maximum price the firm should pay for the project? (Round off to the nearest integer)

Options :

6406536037111. ✖ Rs 12,000

6406536037112. ✖ Rs 15,851

6406536037113. ✔ Rs 19,547

6406536037114. ✖ Rs 22,000

Question Number : 3 Question Id : 6406531863636 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A stock currently trading at Rs 80 can either rise to Rs 100 or fall to Rs 70 in three months. The three-month risk-free rate is 5%. Using the risk-neutral approach, what is the risk-neutral probability that the stock price increases?

Options :

6406536037115. ✖ 53.0%

6406536037116. ✖ 78.6%

6406536037117. ✖ 60.0%

6406536037118. ✔ 46.7%

Question Number : 4 Question Id : 6406531863637 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

An investor opts for a straddle strategy for his portfolio by purchasing a call option and a put option with strike price of Rs 70 and expiration time of 3 months. The call option price is Rs 7 and put option price is Rs 4. What is maximum loss the investor can incur with this strategy?

Options :

6406536037119. ✔ Rs 11

6406536037120. ✖ Rs 3

6406536037121. ✖ Rs 4

6406536037122. ✖ Rs 7

Question Number : 5 Question Id : 6406531863638 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If the interest rate in home country (i_t) is 1% and the interest rate in the foreign country (i_t^*) is 5%,

what is the home country's expected rate of appreciation of the exchange rate according to the uncovered interest parity condition?

$$\left(\frac{E_{t+1}^e - E_t}{E_t} \right)$$

Options :

6406536037123. ✖ 0.94%

6406536037124. ✖ 5.05%

6406536037125. ✔ 3.96%

6406536037126. ✖ 1.94%

Question Number : 6 Question Id : 6406531863639 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A risky asset has an expected rate of return of 18% and a standard deviation of the rate of return of 20%. If the Sharpe ratio of this risky asset is 0.8, then what is the prevailing risk-free rate of return?

Options :

6406536037127. ✖ 1.00%

6406536037128. ✔ 2.00%

6406536037129. ✖ 4.00%

6406536037130. ✖ 5.00%

Question Number : 7 Question Id : 6406531863640 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

In the Black-Scholes pricing formula for a call option, the Delta (Hedge Ratio) is given by $N(d_1)$. If the Delta for a call option is 0.3, and the current stock price is Rs 100, how many whole shares of the stock are needed to perfectly hedge 10 written call options against a small change in the stock price?

Options :

6406536037131. ✔ 3 shares

6406536037132. ✖ 7 shares

6406536037133. ✖ 10 shares

6406536037134. ✖ Insufficient information

Question Number : 8 Question Id : 6406531863641 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A firm is financed with 40% debt and 60% equity. The equity beta is 2.5. The debt has a beta of 0.2. What is the asset beta of the firm?

Options :

6406536037135. ✖ 0.88

6406536037136. ✖ 1.42

6406536037137. ✔ 1.58

6406536037138. ✖ 1.72

Question Number : 9 Question Id : 6406531863642 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Your portfolio consists of two stocks, X and Y, with returns that have a correlation coefficient of 0.2. Stock X has an expected return of 18% and a standard deviation of 30% and Stock Y has an expected return of 10% and a standard deviation of 25%. What is the covariance between the returns of the two stocks?

Options :

6406536037139. ✖ 0.006

6406536037140. ✖ 0.012

6406536037141. ✔ 0.015

6406536037142. ✖ 0.025

Question Number : 10 Question Id : 6406531863643 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You invest 75% of your money in a security with a beta of 1.6, and the rest of your money in a risk-free security. What is the beta of your resulting portfolio?

Options :

6406536037143. ✖ 0.80

6406536037144. ✔ 1.20

6406536037145. ✖ 1.50

6406536037146. ✖ Insufficient information

Question Number : 11 Question Id : 6406531863644 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The current market price of a stock is Rs 500. Next year's expected dividend is Rs 20 per share. The dividend growth rate is expected to be 2% per year forever, and the required rate of return is 6%. According to the Gordon Growth Model, what is the estimated fair price of the stock?

Options :

6406536037147. ✖ Rs 312.5

6406536037148. ✖ Rs 400

6406536037149. ✔ Rs 500

6406536037150. ✖ Rs 625

Question Number : 12 Question Id : 6406531863645 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

An investment is expected to yield a total holding period return of 180% over a period of 25 years. What is the annualized rate of return (in percent) on this investment?

Options :

6406536037151. ✖ 1.13%

6406536037152. ✖ 2.96%

6406536037153. ✔ 4.20%

6406536037154. ✖ 1.40%

Question Number : 13 Question Id : 6406531863646 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A European call option with a strike price of Rs 100 is trading at a price of Rs 5. The stock price is Rs 95, and the risk-free rate is 4% (continuously compounded) for the 1-year time to expiration. According to the put-call parity theorem, what is the corresponding European put option price (in Rupees)?

Options :

6406536037155. ✖ Rs 10.45

6406536037156. ✔ Rs 6.08

6406536037157. ✖ Rs 1.33

6406536037158. ✖ Rs 0.67

Question Number : 14 Question Id : 6406531863647 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The expected return on the market portfolio is 8% and the risk-free rate is 4%. A security has an expected rate of return of 10%. According to CAPM, what is the beta of this security?

Options :

6406536037159. ✖ 0.75

6406536037160. ✖ 1.00

6406536037161. ✔ 1.50

6406536037162. ✖ 2.00

Question Number : 15 Question Id : 6406531863648 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Your expected return on a stock is 14%. The stock has a beta of 1.5. If the risk-free rate is 5% and the expected market return is 10%, what is the stock's alpha (in percent)?

Options :

6406536037163. ✖ -2.0%

6406536037164. ✖ -1.5%

6406536037165. ✔ 1.5%

6406536037166. ✖ 2.0%

Question Number : 16 Question Id : 6406531863649 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The real interest rate in the UK is 2%, and the expected UK inflation rate is 2%. What is the nominal interest rate in the US (in percent)?

Options :

6406536037167. ✖ 3.04%

6406536037168. ✔ 4.04%

6406536037169. ✖ 5.04%

6406536037170. ✖ 6.04%

Question Number : 17 Question Id : 6406531863650 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A long straddle strategy is executed by buying a one-year call option (price Rs 3) and a one year put option (price Rs 5) with the same strike price of Rs 80. If the stock price at expiration is Rs 90, what is the net profit (in Rupees) from this position?

Options :

6406536037171. ✖ Rs 10

6406536037172. ✖ Rs 5

6406536037173. ✔ Rs 2

6406536037174. ✖ There is a loss

Question Number : 18 Question Id : 6406531863651 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A trader executes a strategy by buying a put option on a stock and simultaneously holding a long position (buying) in the same stock. This strategy is known as a

Options :

6406536037175. ✔ Protective Put

6406536037176. ✖ Long Straddle

6406536037177. ✖ Covered Call

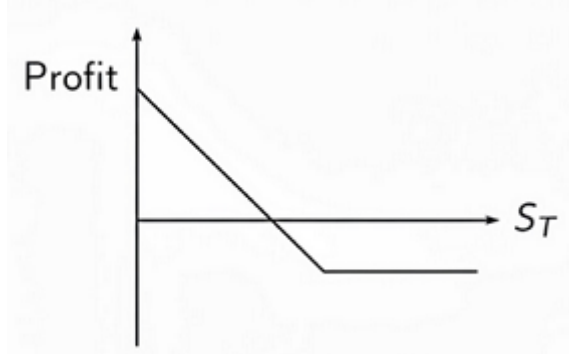
6406536037178. ✖ Long Call

Question Number : 19 Question Id : 6406531863652 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

What option position results in the profit diagram as given in the figure, on the exercise date? S_T denotes the price of the underlying asset and X denotes the strike price.



Options :

6406536037179. ✖ Long call

6406536037180. ✖ Short call

6406536037181. ✔ Long put

6406536037182. ✖ Short put

Question Number : 20 Question Id : 6406531863653 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Kartik's utility function can be written as $U(W) = \sqrt{W}$, where W is his wealth level. He currently has wealth equal to 100 thousand dollars. He invests his wealth in an asset, which will give a return of 10% with probability 0.75, or a return of -10% (that is, a return of negative 10%) with remaining probability by the end of the year (His wealth will not change due to any other reason). This asset gives him as much expected utility as if he invested his wealth in risk-free bonds with an interest rate of $x\%$ per year. What is the value of x ?

Options :

6406536037183. ✖ -1.45%

6406536037184. ✖ 0.00%

6406536037185. ✖ 2.45%

6406536037186. ✔ 4.81%

Question Number : 21 Question Id : 6406531863654 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a stock whose return over the next year is sensitive to one factor – return on an index fund (r_m). Suppose that the sensitivity of the stock return to the index fund is 1.3. If the risk-free interest rate is 4% and risk premium on index fund is 4%, then what is the expected return on the stock over the next year, according to the Arbitrage Pricing Theory?

Options :

6406536037187. ✖ 5.2%

6406536037188. ✖ 2.6%

6406536037189. ✔ 9.2%

6406536037190. ✖ 1.2%

Question Number : 22 Question Id : 6406531863655 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A risky asset has an expected rate of return of 15% and a standard deviation of 25%. If the risk-free rate is 5%, select the correct statements.

Options :

6406536037191. ✓ The Sharpe ratio of the asset is 0.4.

6406536037192. ✓ The risk premium of the asset is 10%.

6406536037193. ✗ The slope of the capital allocation line passing through this asset is 0.8.

6406536037194. ✓ If an investor with utility function $U = \mu - 2\sigma^2$ combines this asset with the risk-free asset, the optimal weight allocated to the risky asset is 0.40.

Question Number : 23 Question Id : 6406531863656 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a put option and a call option on the same stock with the same strike price of Rs 150 and time to maturity of three months. Select the correct statements

Options :

6406536037195. ✗ If the stock price at expiration is Rs 160, call option is out of the money.

6406536037196. ✓ If the stock price at expiration is Rs 160, put option is out of the money.

6406536037197. ✗ If the stock price at expiration is Rs 140, call option is in the money.

6406536037198. ✓ If the stock price at expiration is Rs 140, put option is in the money.

Question Number : 24 Question Id : 6406531863657 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A trader buys two call options on a stock with a strike price of Rs 40 when the call option price is Rs 3. He also buys one put option with the same strike price of Rs 40 and the put option price is Rs 4. The trader makes a profit when the stock price at expiration is

Options :

6406536037199. ✗ Rs 32

6406536037200. ✗ Rs 40

6406536037201. ✓ Rs 48

6406536037202. ✓ Rs 56

Question Number : 25 Question Id : 6406531863658 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A call option on a non-dividend-paying stock has a hedge ratio of 0.4, select the correct statements

Options :

6406536037203. ✓ If the stock price increases by Rs 1, the call option price increases by approximately Rs 0.4.

6406536037204. ✗ If an investor sells 10 call options, she must buy 6 shares of the underlying stock to form a delta-hedged position.

6406536037205. ✓ A put option with same strike price and expiry time has Delta of -0.6.

6406536037206. ✓ The put option price decreases if the stock price increases, as Delta is negative.

Question Number : 26 Question Id : 6406531863659 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

The price of a one-year European call option on a non-dividend-paying stock is Rs 15 (C_0) which has a strike price of Rs 120 (X). The stock price is Rs 115 (S_0), and the annual risk-free rate is 8% (continuously compounded, r). Select the correct statements.

Options :

6406536037207. ✓ A one-year call option with strike price of Rs 125 should have price less than Rs 15.

6406536037208. ✗ A six-month call option with strike price of Rs 120 should have price more than Rs 15.

6406536037209. ✓ The price of a one-year European put option on the same stock is approximately Rs 10.77.

6406536037210. ✓ The sum $P_0 + S_0$ is equal to $C_0 + Xe^{-rT}$.

Question Number : 27 Question Id : 6406531863660 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Security A has an expected return of 12% and a standard deviation of 10%. Security B has an expected return of 18% and a standard deviation of 20%. Suppose that the rates of return of the two securities have a correlation coefficient of 0.6. Based on the given information select the correct statements from below.

Options :

6406536037211. ✓ The covariance of security A and security B is 1.2%.

6406536037212. ✓ If an investor invests 50% of wealth in each security, then his portfolio's expected return is 15%.

6406536037213. ✓ If an investor invests 50% of wealth in each security, then his portfolio's standard deviation is 13.60%.

6406536037214. ✓ If short selling is not allowed, it is not possible to build a portfolio with expected return of 20%.

Question Number : 28 Question Id : 6406531863661 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A money manager calculates that JLK Inc stock has an expected rate of return of 14%. JLK Inc has a beta of 1.5. The risk-free rate of return is 6%, and the expected market rate of return is 12%. Which of the following statements are correct?

Options :

6406536037215. ✓ JLK stock is overpriced.

6406536037216. ✗ JLK stock is underpriced.

6406536037217. ✗ JLK stock would lie above the security market line.

6406536037218. ✓ JLK stock would lie below the security market line.

Question Number : 29 Question Id : 6406531863662 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A stock is trading at Rs 50 today. In 6 months, the stock price can either increase to Rs 60 or decrease to Rs 45. The six-month risk-free rate is 2%. A six-month call option has an exercise price of Rs 55. Select the correct statements based on the given information.

Options :

6406536037219. ✓ The risk-neutral probability that stock price increases is 40%.

6406536037220. ✓ The hedge ratio for the given call option is 1/3.

6406536037221. ✓ The purchase of one share can be hedged with 3 call options.

6406536037222. ✗ According to risk-neutral approach, the price of call option is Rs 7.84.

Question Number : 30 Question Id : 6406531863663 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

OPQ Inc. has an expected return of 15% and a beta of 1.8. The risk-free rate is 6% and the expected return on market portfolio is 12%. Select the correct statements according to CAPM

Options :

6406536037223. ✓ The required expected return for OPQ Inc. should be 16.8%.

6406536037224. ✗ OPQ Inc. stock is underpriced.

6406536037225. ✓ OPQ Inc. has an Alpha of -1.8%.

6406536037226. ✓ An investor following the CAPM recommendation should sell OPQ Inc. shares.

Question Number : 31 Question Id : 6406531863664 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

An investor is considering two assets for her portfolio: a risk-free asset with expected return of 4% and a risky asset with expected return of 10% and standard deviation of 10%. Her utility function is given as $U = \mu - 5\sigma^2$. Given this information select the correct statements.

Options :

6406536037227. ✓ The optimal weight allocated to the risky asset for the investor is 60%.

6406536037228. ✓ The standard deviation of the optimal portfolio is 6%.

6406536037229. ✓ The expected return of the optimal portfolio is 7.6%.

6406536037230. ✓ The Sharpe ratio of the risky asset is 0.6.